

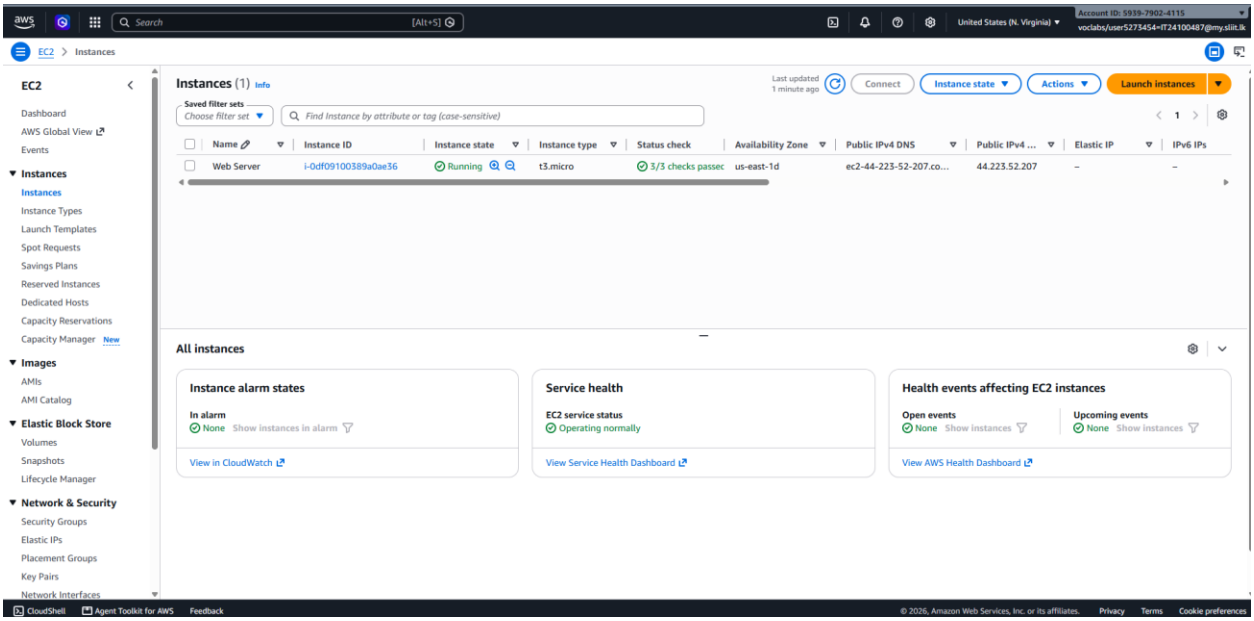
SE3082 – Parallel Computing

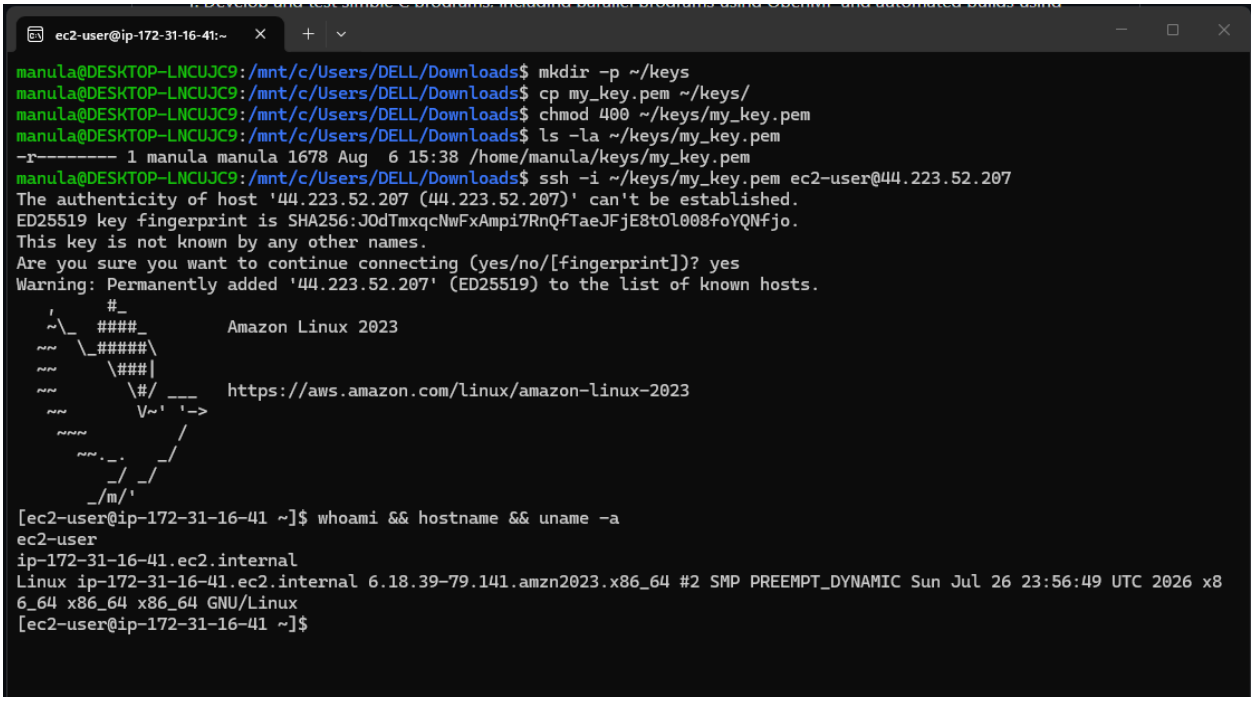
Lab 02

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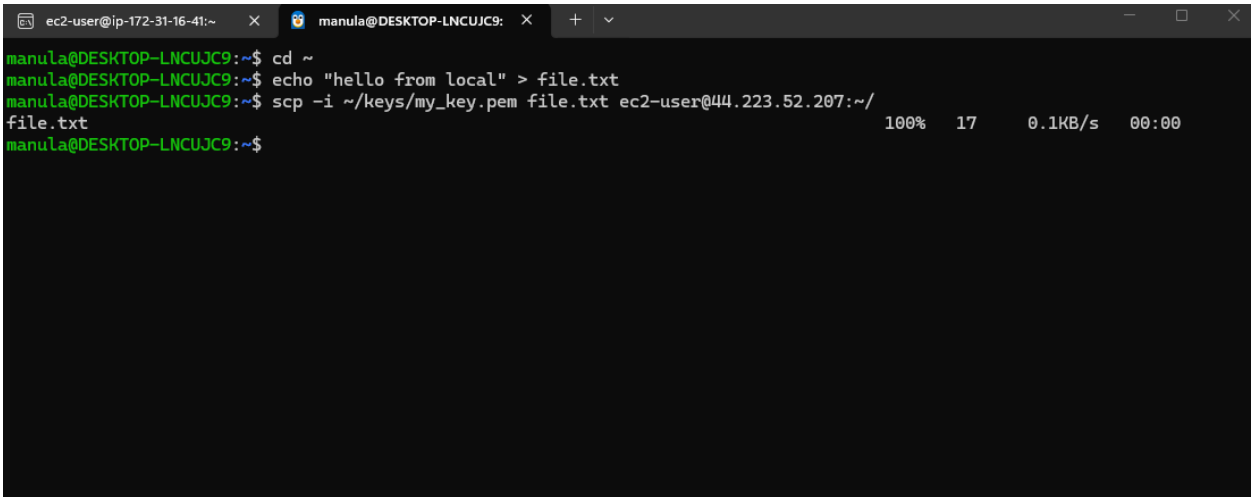
M.K.M. Pehesara

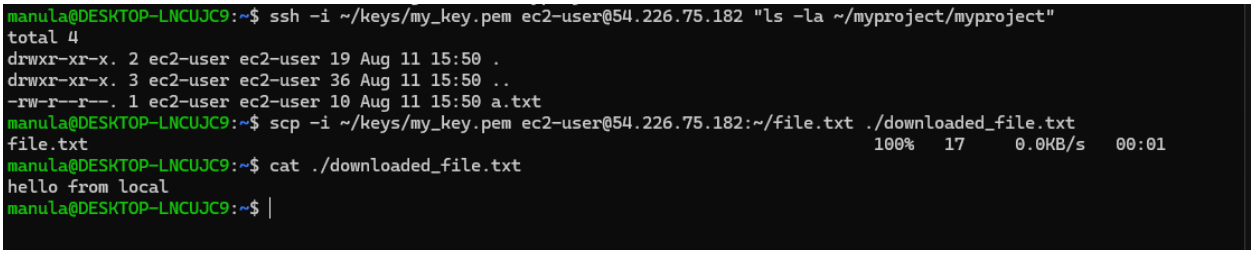
Part 1: Connecting to AWS EC2



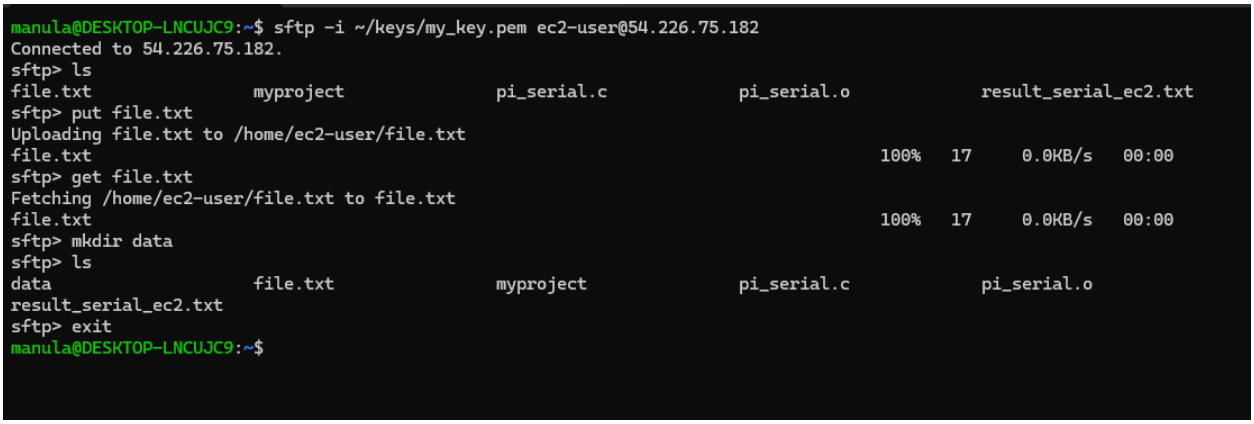


Part 2: File Transfers





Part 3: Using SFTP for File Management



Part 4: Checking Remote Directories without SSHing

```
manula@DESKTOP-LNCUJC9:~$ ssh -i ~/keys/my_key.pem ec2-user@54.226.75.182 "ls -lh /home/ec2-user/"
total 32K
drwxrwxr-x. 2 ec2-user ec2-user   6 Aug 11 16:01 data
-rw-r--r--. 1 ec2-user ec2-user  17 Aug 11 16:01 file.txt
drwxr-xr-x. 3 ec2-user ec2-user  36 Aug 11 15:50 myproject
-rw-r--r--. 1 ec2-user ec2-user 577 Aug  7 06:01 pi_serial.c
-rwxr-xr-x. 1 ec2-user ec2-user 17K Aug  7 06:02 pi_serial.o
-rw-r--r--. 1 ec2-user ec2-user  71 Aug  7 06:03 result_serial_ec2.txt
manula@DESKTOP-LNCUJC9:~$ sftp -i ~/keys/my_key.pem ec2-user@54.226.75.182 <<< $'ls /home/ec2-user/'
Connected to 54.226.75.182.
sftp> ls /home/ec2-user/
/home/ec2-user/data                     /home/ec2-user/file.txt
/home/ec2-user/myproject                /home/ec2-user/pi_serial.c
/home/ec2-user/pi_serial.o              /home/ec2-user/result_serial_ec2.txt
manula@DESKTOP-LNCUJC9:~$
```

Part 5: Automating Transfers

```
manula@DESKTOP-LNCUJC9:~$ mkdir -p ~/localfolder
manula@DESKTOP-LNCUJC9:~$ echo "sync test content" > ~/localfolder/a.txt
manula@DESKTOP-LNCUJC9:~$ rsync -avz -e "ssh -i ~/keys/my_key.pem" ~/localfolder/ ec2-user@54.226.75.182:~/remotefolder/
sending incremental file list
created directory /home/ec2-user/remotefolder
./
a.txt

sent 149 bytes  received 88 bytes  43.09 bytes/sec
total size is 18  speedup is 0.08
manula@DESKTOP-LNCUJC9:~$ rsync -avz -e "ssh -i ~/keys/my_key.pem" ~/localfolder/ ec2-user@54.226.75.182:~/remotefolder/
sending incremental file list

sent 77 bytes  received 12 bytes  16.18 bytes/sec
total size is 18  speedup is 0.20
manula@DESKTOP-LNCUJC9:~$
```

Part 6: OpenMP Programs and timing parallel programs

```
ec2-user@ip-172-31-16-41:~   manula@DESKTOP-LNCUJC9:
manula@DESKTOP-LNCUJC9:~$ ssh -i ~/keys/my_key.pem ec2-user@54.226.75.182
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'
Last login: Fri Aug  7 06:01:51 2026 from 203.94.92.130
[ec2-user@ip-172-31-16-41 ~]$ sudo yum install gcc -y
Last metadata expiration check: 0:41:00 ago on Tue Aug 11 15:29:31 2026.
Package gcc-11.5.0-5.amzn2023.0.5.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-16-41 ~]$ nano parallel_add.c
[ec2-user@ip-172-31-16-41 ~]$ nano parallel_add.c
[ec2-user@ip-172-31-16-41 ~]$ gcc -fopenmp parallel_add.c -o parallel_add
[ec2-user@ip-172-31-16-41 ~]$ ./parallel_add
Running with 4 threads.
Completed computation. Sample result: C[100] = 250.000000
Time taken: 0.000238 seconds
[ec2-user@ip-172-31-16-41 ~]$
```

Part 7: Using make in Linux

